Finance Research Letters 35 (2020) 101293

Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl

Asymmetric volatility in cryptocurrency markets: New evidence from smooth transition GARCH models

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ARTICLE INFO

Keywords:
Cryptocurrencies
Asymmetric volatility
Smooth transition GARCH

J.E.L classification:
G22
C58
G11
G15

ABSTRACT

This paper investigates the presence of asymmetric volatility dynamics in Bitcoin, Ethereum, Ripple, and Litecoin. Asymmetric effects between good and bad news are traditionally modeled using threshold GARCH models that allow only for two possible variance regimes. We experiment a slightly flexible specification for the conditional variance by using a Smooth Transition GARCH (ST-GARCH) model, where a continuum of intermediate states is allowed between the two extreme volatility regimes. We feature an inverted asymmetric reaction for the majority of cryptocurrencies. The presence of positive return-volatility relationship, which is different from other traditional assets, supports the safe-haven hypothesis in cryptocurrencies.

1. Introduction

The presence of asymmetric reaction in the volatility process to unanticipated shocks is a prominent feature of the financial market time series. Known as the leverage effect, the conditional variance of equity returns is often much more affected by negative news compared with positive ones (Black, 1976; Christie, 1982). With the increasing popularity of the digital currencies, there has been a growing literature in recent years analyzing the similarity in the volatility properties between cryptocurrencies and other financial assets (see e.g. Baur et al., 2018; Dyhrberg, 2016a; Bouri et al., 2017a; and Klein et al., 2018). Shedding further light on the potential asymmetric behavior in the cryptocurrency markets is of particular importance to check whether they would offer opportunities in terms of portfolio diversification and risk management. More specifically, a positive return-volatility relationship, which is different from the other traditional financial assets, may point toward possible hedge and safe-haven properties of cryptocurrencies.[1] Bitcoin, as the most popular and traded cryptocurrency, has experienced extreme fluctuation since its introduction in 2009. Large occasional price swings, such as the market crash of December 2013 and the unprecedented price levels in late 2017, point to the need for modeling the conditional variance of virtual currencies, and whether asymmetric behavior does exist (see Fig. 1).

Asymmetry in the volatility process is traditionally detected using asymmetric Generalized Autoregressive Conditional Heteroscedasticity (GARCH) models such as exponential GARCH (EGARCH) model of Nelson (1991) and threshold GJR-GARCH

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[1] Hedging and safe haven abilities are the prominent features of gold market. The volatility of the gold is found to exhibit an inverted asymmetric volatility pattern, as positive return shocks lead to increased volatility (see e.g. Baur, 2012).

https://doi.org/10.1016/j.frl.2019.09.008
Received 18 February 2019; Received in revised form 2 September 2019; Accepted 9 September 2019
Available online 10 September 2019
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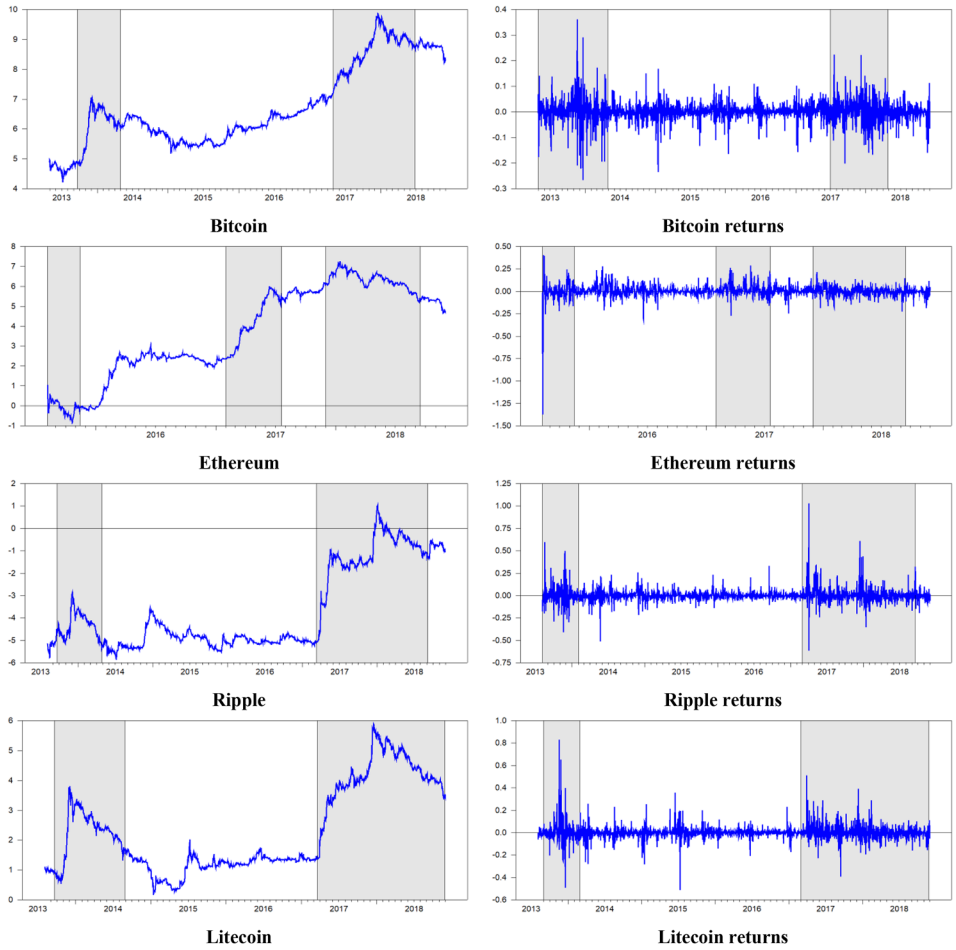


Fig. 1. Daily closing prices and returns for four major cryptocurrencies (US Dollars)
Notes: Plots corresponds to the natural logarithm of closing prices and their returns in the major cryptocurrency markets. Data cover the period from April 28, 2013 to December 1, 2018, except for Ethereum for which data starts from August 07, 2015.

model of Glosten et al. (1993). Some recent studies have implemented different GARCH-type models to examine the volatility properties of the cryptocurrency markets.[2] For instance, Katsiampa (2017) has proposed to compare the modeling performance of six variance models to estimate the volatility of Bitcoin returns, among them the Asymmetric Power ARCH (APARCH) model of Ding et al. (1993) and the Component GARCH (CGARCH) model of Engle and Lee (1999).[3] The asymmetric GARCH models that have been proposed to capture the leverage effect allow only for two possible regimes, i.e., low volatility regime and high volatility regime, where the transition across regimes is abrupt. As more than two-variance regimes would potentially characterize financial times series, it would be interesting to experiment a slightly flexible specification for the conditional volatility of the virtual currencies.

In this paper, we propose the implementation of the class of Smooth Transition GARCH (ST-GARCH) models, where the presence of a continuum of intermediate states are allowed between the two extreme regimes of volatilities (see González-Rivera, 1998; Hagerud, 1996). The ST-GARCH is a generalization of the GJR-GARCH model, and the introduction of a smooth-transition mechanism in the conditional variance equation offers greater flexibility as the transition from a low- to a high-volatility regime could be gradual. To shed further light on the possible asymmetric volatility in the cryptocurrencies, the ST-GARCH specification is estimated for the most popular ones in circulation today, and with highest market capitalization, namely, Bitcoin, Ethereum, Ripple and Litecoin. To the best of our knowledge, no other study has investigated the asymmetric volatility phenomenon for the digital currencies by introducing a smooth-transition mechanism in the GARCH process.

[2] For the implementation of EGARCH and GJR-GARCH models to cryptocurrency markets see e.g. Baur and Dimpfl (2018), Dyhrberg (2016b), and Bouri et al. (2017b).

[3] In their replication of the study of Katsiampa (2017), Charles and Darné (2019) have reported that the asymmetric parameters are not statistically significant for all the asymmetric GARCH models.

The remainder of the paper is organized as follows: Section 2 describes the times series proprieties of our major virtual currencies. In Section 3, we present the econometric framework. Section 4 discusses the main empirical results. Finally, concluding remarks are summarized in Section 5.

2. Data

The dataset we consider in our empirical exercise consists of daily closing prices in US Dollars, which are collected from CoinMarketCap (https://coinmarketcap.com/coins/).[4] We focus on the four most representative cryptocurrencies in terms of diffusion and market capitalization: Bitcoin, Ethereum, Ripple, and Litecoin. The returns are calculated by taking the natural logarithm of the ratio of two consecutive prices. Our data cover the period from April 28, 2013, to December 1, 2018, except for Ethereum for which data starts from August 07, 2015.

The summary statistics of daily closing returns of our four cryptocurrencies are reported in Table 1. For comparison purposes, we consider the same daily period from August 07, 2015 to December 1, 2018, to examine their statistical properties. The highest average return is found in Ethereum and Ripple, while Bitcoin exhibits the lowest variability among the major cryptocurrency markets. The return distribution is positively skewed for Ripple and Litecoin, but skewed left for Bitcoin and Ethereum. Also, all of the returns are heavy-tailed with kurtosis more significant than a normal distribution. As shown in Table 1, the Jarque-Bera test confirms the leptokurtic behavior of our digital currencies. Fig. 2 plots the histograms of the daily closing returns for Bitcoin, Ethereum, Ripple, and Litecoin. While series are more or less symmetrically distributed, it is clear that they are more peaked than a normal distribution. The Ljung and Box (1978) test reveal substantial evidence of residual autocorrelation for all the returns.[5] Results from Engle (1982)'s Lagrange Multiplier (LM) ARCH test for conditional heteroskedasticity confirms the presence of ARCH effects. This justifies the usefulness of a GARCH framework to model the conditional volatility. Finally, we have conducted Augmented Dickey-Fuller (ADF) unit-root tests which confirm the stationarity of the return series.

3. Econometric approach

In order to capture the presence of a potential asymmetric behavior in conditional volatility, we propose to apply an ST-GARCH specification for the cryptocurrency returns. Contrary to the standard asymmetric GARCH models, ST-GARCH model allows for several intermediate regimes (more than two) in conditional variance. The ST-GARCH model is a generalization of the GJR-GARCH model (Glosten et al., 1993) where the transition from a low- to a high-volatility regime could be smooth (see González-Rivera, 1998; Hagerud, 1996).

Considering a GARCH process u_t having the form $u_t = \varepsilon_t \sqrt{h_t}$, where h_t its conditional variance at time t and ε_t iid $N(0,\ 1)$. A smooth-transition GARCH($p,\ q,\ d$) can be written as:

$$h_t = \omega + \sum_{i=1}^{p} \alpha_{1i} u_{t-i}^2 + \left(\sum_{i=1}^{p} \alpha_{2i} u_{t-i}^2 \right) F(u_{t-d},\ \gamma) + \sum_{i=1}^{q} \beta_i h_{t-i} \tag{1}$$

where $F(u_{t-d},\ \gamma)$ is the transition function having a logistic form as follows:

$$F(u_{t-d},\ \gamma) = [1 + \exp(-\gamma u_{t-d})]^{-1} - \frac{1}{2},\ \gamma > 0 \tag{2}$$

where u_{t-d} is the transition variable, $d \leq p$, and γ is the slope parameter determining the speed of transition across regimes. The logistic function is bounded $-1/2 < F(u_{t-d},\ \gamma) < 1/2$. For strictly positive conditional variances, sufficient conditions are $\omega > 0$, $\alpha_{1i} \geq 0$, $\alpha_{1i} \geq \frac{1}{2}|\alpha_{2i}|$ for $i = 1,\ \ldots,\ p$, and $\beta_i > 0$ for $i = 1,\ \ldots,\ q$.

The logistic transition function is appropriate for a process where the dynamic of the conditional variance differs depending on the sign of the innovation. Then, capturing the asymmetric behavior of conditional variances to negative news versus positive news within a logistic ST-GARCH is very appealing. When $u_{t-d} \to\to -\infty$, the logistic function will be equal to $-1/2$, while if $u_{t-d} \to\to +\infty$, the transition function will correspond to $1/2$. The ST-GARCH model offers the possibility of intermediate regimes where the moving among regimes is gradual.[6] Within this framework, if the leverage effect exists, i.e., negative innovations tend to produce higher variance than positive ones, then the parameter α_{2i} is expected to be positive.

In line with Bollerslev (1986), the necessary and sufficient condition for the second-order stationarity is derived for each regime. For the upper regime, i.e., when $u_{t-d} \to\to -\infty$ and $F(u_{t-d},\ \gamma) = 1/2$, the process is covariance stationary if and only if:

$$\sum_{i=1}^{p} \alpha_{1i} + \frac{1}{2} \sum_{i=1}^{p} \alpha_{2i} + \sum_{i=1}^{q} \beta_i < 1 \tag{3}$$

When $u_{t-d} \to\to +\infty$, i.e., $F(,\ u_{t-d},\ \gamma) = -1/2$, the condition for covariance stationarity in the lower regime is:

[4] As cryptocurrency markets are open 24 h a day and seven days a week, then for the closing values we consider prices at midnight (UTC).

[5] For Ljung–Box test and ARCH test, statistics reported in Table 1 corresponds to the lowest p-value of lags 1–10.

[6] In the threshold GARCH model introduced by Glosten et al. (1993), only two volatility regimes (a low volatility regime and a high volatility regime) are allowed with respect to lagged positive and negative innovations, where the transition across the two variance regimes is rather abrupt.

Table 1
Summary statistics.

	Bitcoin	Ethereum	Ripple	Litecoin
Mean	0.220%	0.307%	0.309%	0.174%
Std. Dev.	0.038	0.079	0.076	0.057
Min.	−20.207%	−137.399%	−60.172%	−39.515%
Max.	22.351%	40.346%	101.097%	51.034%
Skewness	−0.277	−3.437	3.066	1.278
	(0.000)	(0.000)	(0.000)	(0.000)
Kurtosis	4.750	64.979	38.979	12.938
	(0.000)	(0.000)	(0.000)	(0.000)
Jarque-Bera	1156.101	215,615.695	78,695.277	8791.447
	(0.000)	(0.000)	(0.000)	(0.000)
Ljung–Box (10)	25.267	19.265	39.059	34.885
	(0.001)	(0.007)	(0.000)	(0.000)
ARCH (10)	240.968	121.140	187.320	209.565
	(0.000)	(0.000)	(0.000)	(0.000)
ADF Test Statistics	−24.998	−20.655	−17.122	−23.657
	(0.000)	(0.000)	(0.000)	(0.000)

Notes: Data span the daily period from August 07, 2015 to December 1, 2018. Std. Dev., Min., and Max. stands for standard deviation, minimum, and maximum. For the Jarque-Bera test, the null hypothesis is of normality. We also run the Ljung–Box test and ARCH Engle's test up to 10 lags, where the null assumptions are no autocorrelation and no ARCH effect, respectively. Statistics reported here corresponds to the lowest *p*-value of lags 1–10. For the Augmented Dickey–Fuller (ADF) test, the null hypothesis is that the series has a unit root.

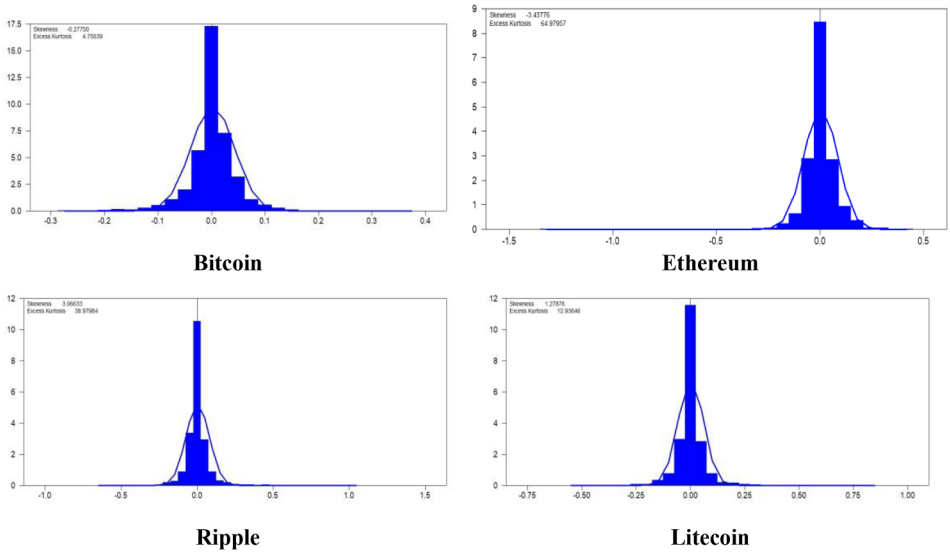


Fig. 2. The histogram of the returns in the major cryptocurrency markets
Note: The histogram of the return distribution in the cryptocurrency markets. Data cover the daily period from August 07, 2015 to December 1, 2018.

$$\sum_{i=1}^{p} \alpha_{1i} - \frac{1}{2}\sum_{i=1}^{p} \alpha_{2i} + \sum_{i=1}^{q} \beta_i < 1 \tag{4}$$

In the intermediate regimes, as in the case of the mid-regime when $u_{t-d} = 0$ and $F(0, \gamma) = 0$, the covariance process is stationary if and only if:

$$\sum_{i=1}^{p} \alpha_{1i} + \sum_{i=1}^{q} \beta_i < 1 \tag{5}$$

Our ST-GARCH Eq. (1) is estimated for the four significant cryptocurrencies, namely, Bitcoin, Ethereum, Ripple, and Litecoin, using quasi-maximum-likelihood (QML) procedure.[7] It is worth highlighting that the use of QML approach for smooth-transition-type

[7] QML estimates are obtained using the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm available within RATS software package (RATS 9.2). As discussed in Bollerslev and Wooldridge (1992), the QML estimators are generally consistent and have a limiting normal distribution when the normality is violated. See also Chan and McAleer (2003) for a discussion of the robustness of the QML for smooth-transition-type models.

Table 2
The different flavors of estimated GARCH models.

GARCH	$h_t = \omega + \alpha_1 u_{t-i}^2 + \beta h_{t-1}$
EGARCH	$\ln h_t = \omega + \alpha_1 \lvert u_{t-1} \rvert / \sqrt{h_{t-1}} + \alpha_2 u_{t-1} / \sqrt{h_{t-1}} + \beta \ln h_{t-1}$
GJR-GARCH	$h_t = \omega + \alpha_1 u_{t-i}^2 + \alpha_2 u_{t-i}^2 I(u_{t-1} < 1) + \beta h_{t-1}$
ZARCH	$h_t^{1/2} = \omega + \alpha_1 \lvert u_{t-1} \rvert + \alpha_2 I(u_{t-1} < 1) \lvert u_{t-1} \rvert + \beta h_{t-1}^{1/2}$
ST-GARCH	$h_t = \omega + \alpha_1 u_{t-i}^2 + \alpha_2 u_{t-i}^2 F(u_{t-d}, \gamma) + \beta h_{t-1}$

Note: All models are with one lag of the innovation and one lag of volatility ($p = 1, q = 1$).

models is possibly sensitive to the presence of extreme observations and outliers (see, e.g., Chan and McAleer, 2003).[8] More specifically, the estimation of transition parameter γ is subject to computational difficulties as conventional optimization algorithms do not seem to perform well in locating the optimum of the associated likelihood function. In our estimation of our ST-GARCH models, we have followed the approach of Chan and Theoharakis (2011) by using a simple re-parameterization of the transition parameter with $\lambda = 1/\sqrt{\gamma}$. The basic idea is to transform the parameter space so that the log-likelihood function would have a much smoother surface and therefore alleviate the numerical problem. The implementation of the transform has a significant effect on mitigating outliers, providing a set of estimates that are much more robust than the original parameterization.

4. Empirical results

For comparison purposes, different GARCH-type models have been estimated in addition to ST-GARCH model. As shown in Table 2, five variance models have examined: (1) standard GARCH model of Bollerslev (1986); (2) exponential GARCH (EGARCH) model of Nelson (1991); (3) threshold GJR-GARCH model of Glosten et al. (1993); (4) threshold GARCH (ZARCH) model of Zakoian (1994); and (5) ST-GARCH model.[9] The best model for p and q is selected using different information criteria, namely the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Low orders of GARCH-type models are found to best fits our cryptocurrency returns series (all models with one lag of the innovation and one lag of volatility).[10] Once the different GARCH-type models are estimated, diagnostic tests for the squared residuals are conducted to evaluate the goodness of fit of the selected models. Also, to discriminate among the various estimated GARCH models, information selection criteria are performed, namely AIC and BIC, where the smaller the values correspond to a better fit.

Estimation results for the major cryptocurrencies using different GARCH-type models are reported from Tables 3 to 6. Comparing the log-likelihood values suggests that an ST-GARCH model provides the closest fit to the empirical processes for Bitcoin, Ripple, and Litecoin. Similarly, according to the information criteria (AIC and BIC), the goodness of fit of the smooth-transition model compared to the other GARCH-type models is confirmed, except for Ethereum, where a ZARCH model seems to be the best specification.[11] Also, diagnostic tests of McLeod and Li (1983) and Engle (1982) for the presence of ARCH effect disclose no conditional heteroscedasticity in the squared standardized residuals in ST-GARCH models.[12]

Concerning coefficient estimates, there is a high volatility persistence in the major cryptocurrencies illustrated by the sum of the ARCH and GARCH terms ($\alpha_1 + \alpha_2/2 + \beta$) which is close to 1. The coefficients α_1 and α_2 are relatively weak, while β is relatively high, as evidence of volatility clustering. As confirmed by Fig. 1, periods of higher variability are followed by periods of relative quietness. Then empirical volatilities tend to cluster. Furthermore, ST-GARCH specification reveals that the coefficient for the asymmetric term α_2 is negative and highly significant for three digital currencies, namely Bitcoin, Ripple, and Litecoin. A negative coefficient implies the existence of a positive return-volatility relationship in the cryptocurrency markets, implying that volatility tends to rise in response to goods news and fall in response to bad news. In other words, during periods of positive shocks ($u_{t-1} > 0$), cryptocurrency markets seem to exhibit larger volatility; however, after a negative return shock ($u_{t-1} < 0$), there is lower volatility in the digital currency markets. This outcome contrasts the so-called leverage effect as good news seems to have more impact on volatility than the bad news.

[8] As shown by Chaim and Laurini (2018), Charles and Darné (2019), Scaillet et al. (2018), Trucios (2019) and Catania and Grassi (2017) cryptocurrencies are subject to a number of drastic shocks (called large shocks, outliers or jumps). These shocks may pose difficulties for the identification and estimation of GARCH models governing the conditional volatility of returns.

[9] In the threshold GARCH model Zakoian (1994), the functional form is quite similar to the GJR-GARCH model, but instead of modeling the conditional variance, Zakoian (1994) models the conditional standard deviation.

[10] It is possible to introduce a lagged return in the mean equation by estimating GARCH-type models with AR(1) process for the mean as in Katsiampa (2017). The results remain quite similar and available from authors upon request.

[11] For the sake of harmonization, our different GARCH-type models have been re-estimated for Bitcoin, Ethereum, Ripple and Litecoin using the same time period as for Ethereum, i.e. from August 07, 2015 to December 1, 2018. The obtained results were quite similar as the ST-GARCH model seems to be again the best specification. To save space, the results are not reported here, but are available in a supplementary Appendix online.

[12] McLeod and Li (1983) proposed a formal test for ARCH effect based on the Ljung–Box procedure by looking at the serial correlation in the squared residuals.

Table 3
Estimation results of conditional variance for Bitcoin.

	GARCH	EGARCH	GRJ-GARCH	ZARCH	ST-GARCH
Constant	0.001	0.001***	0.001*	0.002***	0.001**
	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)
ω	0.000***	−0.526***	0.000***	0.001***	0.000***
	(0.000)	(0.084)	(0.000)	(0.000)	(0.000)
α_1	0.140***	0.309***	0.150***	0.193***	0.136***
	(0.019)	(0.035)	(0.023)	(0.016)	(0.015)
β	0.850***	0.950***	0.853***	0.843***	0.852***
	(0.017)	(0.001)	(0.017)	(0.016)	(0.015)
α_2		0.016***	−0.028	−0.035***	−0.038**
		(0.000)	(0.019)	(0.014)	(0.020)
γ					6.950*
					4.154
Log likelihood	4033.361	3850.868	4034.489	3901.507	**4040.960**
AIC	−3.945	−3.765	−3.945	−4.033	**−4.054**
BIC	−3.934	−3.751	−3.931	−4.017	**−3.038**
$Q^2(10)$	6.268	126.803	10.167	8.001	2.247
	(0.508)	(0.000)	(0.179)	(0.238)	(0.325)
ARCH(10)	5.947	83.205	7.408	7.032	2.002
	(0.745)	(0.000)	(0.271)	(0.633)	(0.571)

Notes: The different GARCH-type models are estimated over the daily period from April 28, 2013 to December 1, 2018. Standard errors of estimates are reported in parentheses. ***, **, and * denote 1%, 5%, and 10% significance levels, respectively. $Q^2(10)$ is McLeod and Li (1983) test and *ARCH* (10) is Engle's LM ARCH test applied to the squared standardized residuals up to 10 lags. The *p*-values associated with the statistical tests are presented in brackets. The bold values correspond to the highest value of the log-likelihood value and the lowest values of the AIC and BIC among the different GARCH-type models.

This inverted asymmetric reaction has been revealed by Baur et al. (2018) who detected a significant difference in the volatility processes between Bitcoin and other assets (gold, US dollar, and equity markets). Using the GJR model, the authors found that Bitcoin becomes more volatile following unanticipated positive shocks.[13] Similarly, evidence of a positive relationship between return shocks and volatility has been provided by Bouri et al. (2017b) during the pre-crash period of Bitcoin, i.e., before December 2013. An inverse leverage effect is considered as a prominent feature of Gold (see, e.g., Baur, 2012). In line with the hedge and safe haven properties, investors would buy gold and transmit the volatility and uncertainty to the gold market as the price of gold increases during episodes of financial turmoil. However, enhancing financial conditions, uncertainty, and volatility would decrease in the gold market as the price of gold falls.[14]The similarities between gold and major cryptocurrencies have been highlighted in Bouri et al. (2017a) and Dyhrberg (2016b), among others.[15] Our empirical results appear supportive of the hedging and safe-haven capabilities of digital currencies. We note that for the case of Ethereum, there is no asymmetry detected across our different GARCH-type models, as shown in Table 4. The relatively short period since the introduction of Ethereum is not sufficient to render the asymmetric behavior possible or robust.

It is worth highlighting that an ST-GARCH specification provides further flexibility than variance models with only two volatility regimes. As shown in Tables 3–6, there is considerable variability in the speed of transition between the extreme variance regimes across the different ST-GARCH models. The transition parameter γ is ranging from 6.95 for Bitcoin to 135.88 for Ripple. The transition from a low- to a high-volatility regime is smoother for Bitcoin, rejecting the threshold specification. While for Ripple where the transition is sharp, a threshold GJR model seems to have a quite closer fit to ST-GARCH specification as displayed in Table 5. The introduction of a smooth-transition mechanism in the GARCH specification offers extra flexibility in the transition process as the GJR-GARCH model is included as a particular case. The use of ST-GARCH is very appealing in capturing the asymmetric volatility as the smoothness of the transition across variance regimes is determined endogenously from the data.

Finally, the size of market capitalization would influence the volatility pattern of our significant crypto assets.[16] With small market value, the prices of alternative virtual currencies (altcoins) fluctuate considerably wider than Bitcoin as the most dominant

[13] Baur et al. (2018) has replicated the study of Dyhrberg (2016a) who did not find asymmetric reaction to positive and negative shocks using EGARCH model.

[14] For a literature on Gold as a hedge and a safe haven against assets such as stocks, bonds, and US Dollar, see e.g. Baur and Lucey (2010), Baur and McDermott (2010), Capie et al. (2005), among others.

[15] The inverted asymmetric volatility effect is also a typical finding for commodities. To the extent that speculators are attracted by increasing prices, this will lead to further higher prices and increasing volatility. Another explanation is related to is the role of inventory levels and storage. Low inventory is likely to render future prices more volatile due to the uncertainty regarding the future supply of the commodity. As the supply of digital currencies is limited by the design of the protocol, the inventory level hypothesis seems to be also consistent with the inverted asymmetry in the cryptocurrency markets.

[16] Based on data from coinmarketcap.com recorded in December 1, 2018, Bitcoin has the highest market share which captures around 53.85% of the total market. Ripple and Litecoin represent respectively 10.85% and 1.49% of the total cryptocurrency market value.

Table 4
Estimation results of conditional variance for Ethereum.

	GARCH	EGARCH	GRJ-GARCH	ZARCH	ST-GARCH
Constant	0.000	0.012***	0.000	0.003*	0.001
	(0.002)	(0.000)	(0.002)	(0.002)	(0.002)
ω	0.000***	−2.552	0.000***	0.005***	0.001***
	(0.000)	(0.004)	0.000)	(0.001)	(0.000)
α_1	0.313***	0.343***	0.313***	0.259***	0.330***
	(0.094)	(0.008)	(0.066)	(0.027)	(0.039)
β	0.655***	0.569***	0.655***	0.761***	0.620***
	(0.049)	(0.007)	(0.049)	(0.025)	(0.038)
α_2		0.091	0.002	0.051	0.460
		(0.111)	(0.069)	(0.106)	(0.309)
γ					8.148 (6.591)
Log likelihood	1526.713	1438.291	1526.714	**1539.178**	1528.294
AIC	−2.513	−2.365	−2.511	**−2.667**	−2.512
BIC	−2.496	−2.344	−2.490	**−2.640**	−2.487
$Q^2(10)$	3.220	14.909	3.219	7.112	3.030
	0.570	0.135	0.660	0.714	0.601
ARCH(10)	4.715	35.863	4.718	5.153	3.177
	0.909	0.000	0.909	0.880	0.976

Notes: The different GARCH-type models are estimated over the daily period from August 07, 2015 to December 1, 2018. Standard errors of estimates are reported in parentheses. ***, **, and * denote 1%, 5%, and 10% significance levels, respectively. $Q^2(10)$ is McLeod and Li (1983) test and *ARCH*(10) is Engle's LM ARCH test applied to the squared standardized residuals up to 10 lags. The *p*-values associated with the statistical tests are presented in brackets. The bold values correspond to the highest value of the log-likelihood value and the lowest values of the AIC and BIC among the different GARCH-type models.

Table 5
Estimation results of conditional variance for Ripple.

	GARCH	EGARCH	GRJ-GARCH	ZARCH	ST-GARCH
Constant	−0.004***	−0.002*	−0.002***	−0.000	−0.002***
	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)
ω	0.001***	−1.819***	0.001***	0.012***	0.001***
	(0.000)	(0.031)	(0.000)	(0.001)	(0.000)
α_1	0.495***	0.305***	0.244***	0.357***	0.285***
	(0.063)	(0.026)	(0.122)	(0.063)	(0.074)
β	0.537***	0.757***	0.698***	0.544***	0.695***
	(0.038)	(0.007)	(0.036)	(0.031)	(0.035)
α_2		0.152***	−0.136***	−0.171***	−0.148***
		(0.008)	(0.046)	(0.056)	(0.116)
γ					135.878* (81.309)
Log likelihood	2748.186	2734.267	2754.253	2739.844	**2756.573**
AIC	−2.832	−2.806	−2.837	−2.834	**−2.851**
BIC	−2.821	−2.792	−2.823	−2.821	**−2.837**
$Q^2(10)$	4.799	8.147	3.905	1.846	3.948
	(0.904)	(0.614)	(0.951)	(0.997)	(0.949)
ARCH(10)	4.715	8.279	3.874	2.916	3.903
	(0.909)	(0.601)	(0.952)	(0.997)	(0.951)

Notes: The different GARCH-type models are estimated over the daily period from April 28, 2013 to December 1, 2018. Standard errors of estimates are reported in parentheses. ***, **, and * denote 1%, 5%, and 10% significance levels, respectively. $Q^2(10)$ is McLeod and Li (1983) test and *ARCH*(10) is Engle's LM ARCH test applied to the squared standardized residuals up to 10 lags. The *p*-values associated with the statistical tests are presented in brackets. The bold values correspond to the highest value of the log-likelihood value and the lowest values of the AIC and BIC among the different GARCH-type models.

virtual currency. Even a handful of trading volume makes altcoins more vulnerable to price manipulation (see, e.g., Phillip et al., 2018). As shown in Table 1, Ripple and Litecoin exhibit huge volatility either measured by the standard deviation or the return range (max-min). Their higher vulnerability to the behavior of market participants would entail a sharper transition mechanism from a low- to a high volatility regime. However, because of its dominant market position, Bitcoin is expected to have a more gradual switching behavior.

5. Conclusion

In this paper, we have investigated the asymmetric volatility dynamic in the major cryptocurrency markets. We have implemented an ST-GARCH specification which offers greater flexibility in capturing the differential impact of positive shocks versus

Table 6
Estimation results of conditional variance for Litecoin.

	GARCH	EGARCH	GRJ-GARCH	ZARCH	ST-GARCH
Constant	−0.002	0.006***	−0.001	0.002*	0.000
	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
ω	0.000***	−0.536***	0.000***	0.003***	0.000***
	(0.000	(0.012)	(0.000)	(0.000)	(0.000)
α_1	0.094***	0.240***	0.121***	0.152***	0.058***
	(0.010)	(0.001)	(0.015)	(0.013)	(0.008)
β	0.885***	0.575***	0.888***	0.902***	0.897***
	(0.010)	(0.001)	(0.010)	(0.008)	(0.010)
α_2		0.016***	−0.066***	−0.047***	−0.130***
		(0.000)	(0.015)	(0.015)	(0.015)
γ					15.417** (12.270)
Log likelihood	2843.554	2666.225	2853.854	2829.256	**2891.509**
AIC	−2.920	−2.736	−2.929	−2.924	**−2.967**
BIC	−2.908	−2.722	−2.915	−2.917	**−2.950**
$Q^2(10)$	0.987	52.789	0.833	2.304	1.346
	(0.999)	(0.000)	(0.999)	(0.993)	(0.999)
ARCH(10)	0.985	47.882	0.839	2.211	1.343
	(0.999)	(0.000)	(0.999)	(0.994)	(0.999)

Notes: The different GARCH-type models are estimated over the daily period from April 28, 2013 to December 1, 2018. Standard errors of estimates are reported in parentheses. ***, **, and * denote 1%, 5%, and 10% significance levels, respectively. $Q^2(10)$ is McLeod and Li (1983) test and *ARCH* (10) is Engle's LM ARCH test applied to the squared standardized residuals up to 10 lags. The *p*-values associated with the statistical tests are presented in brackets. The bold values correspond to the highest value of the log-likelihood value and the lowest values of the AIC and BIC among the different GARCH-type models.

adverse shocks. Our estimation results suggest a satisfactory modeling performance of ST-GARCH models in terms of log-likelihood and information criteria compared to other conditional variance models. We also found strong evidence of an inverted asymmetric reaction for the majority of digital currencies; that is, good news seems to have more impact on volatility than bad news. This atypical asymmetric effect is a common feature encountered in safe-haven assets such as gold. In times of financial turmoil, investors would transmit volatility and uncertainty to cryptocurrency markets in their quest for a hedge against a distressed equity market.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.frl.2019.09.008.

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